Desk Number ________

Student Number ________________________

Student Name ________________________

School of Science and Engineering

MIDTERM EXAMINATION

Semester 1, 2016

CSC1001 Introduction to Computer Science

Examination Duration: 120 minutes

Reading Time: 10 minutes

This examination has __3__ questions.

Exam Conditions:

This is a FORMAL Examination

This is a RESTRICTED OPEN BOOK Exam. Maximum of one (1) sheet of handwritten notes double sided are permitted

Materials Permitted In The Exam Venue:

Maximum of one (1) sheet of handwritten notes double sided are permitted. **NO OTHER MATERIALS PERMITTED**

Any calculators without the functionalities of programming and file storage are permitted.

Materials To Be Supplied To Students:

1 × 12 Page Answer Booklet

Question 1. (10 × 3% = 30%)

Pick the correct option in each of the following sub-questions. Note that only ONE option is correct.

1.1) Which of the following is NOT a part of the Von Neumann architecture?

A. central processing unit B. main memory C. input device D. graphical processing unit

1.2) Assume that before time = 0, the voltage level is positive. Then the signals shown in the following figure can be translated into NRZI code

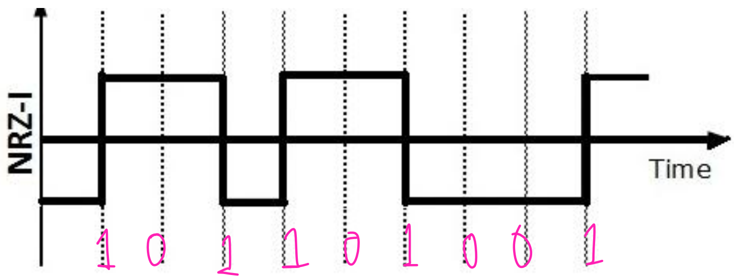

Figure 1

A. 1101101001 B. 0110110001 C. 1001001110 D. 0101101001

1.3) Which of the following is NOT a high-level programming language?

A. C language B. Java language C. Assembly language D. C# language

1.4) Concerning programming languages, which of the following is correct?

A. High level languages can be executed without translated into low-level languages;

B. High level languages have higher language efficiency

C. High level languages have higher development efficiency

D. Assembly language is also called machine language

1.5) Binary number 100100.111 equals to decimal number

A.36.875 B.36.825 C.36.175 D.36.5

1011 1100 10001111

1.6) Hexadecimal number BC.8F equals to binary number

A.11111100.10001110

B.10111101.10001111

C.10111100.10001111

D.10111100.11001101

1.7) Concerning information unit, which of the following statement is incorrect?

A. Bit is the smallest information unit in computer programming;

B. 1 MB = 2^20 bit;

C. 1 GB = 1024 MB;

D. 1 byte = 8 bit;

1.8) Concerning Python variables, which of the following is incorrect?

A. Variable names must start with a letter or an underscore;

B. Variable names can only contain letters, numbers and underscore;

C. Variable names are case insensitive;

D. A variable is a named space in the memory;

1.9) What is the output of the following program?

```
x=10
y=15.5
z=x+y**2*10/(10-20)
print(z)
```

10 -240.25

A. 230.25 B. 200.25 C. -230.5 D. -230.25

1.10) What is the output of the following program?

```
>>> a,b=divmod(104,25)
>>> a,b
```

A. (4,4) B. (4,0) C. (5,21) D. (5,0)

Question 2. (10 × 4% = 40%)

Pick the correct option/s in each of the following sub-questions. Note that there may be MULTIPLE correct options for each sub-question.

2.1) Concerning Python data types, which of the following statements is incorrect?

A. Some operations are prohibited on certain types;

B. The data type of an object can be checked using type() function;

C. `myStr = 'abc' + 1` is a correct assignment statement;

D. The type of a variable will be fixed during program execution once the variable is created;

2.2) Concerning the following program, which of the following statements are correct?

```
a = int('123')
print(a)
b = float('35.7')
print(b)
c = int('123.ab')
print(c)
```

A. All three print() statements will output a number;

B. After the program has been executed, variable a is of integer type;

C. After the program has been executed, variable c is of integer type;

D. Function float() is used for data type conversion;

2.3) The following 4 options show 4 statements and their outputs; in which option the output is incorrect?

A.
```
>>> str(100.5)
'100.5'
```

B.
```
>>> str(1+10+100)
'1+10+100'
```

C.
```
>>> str("Hello world")
'Hello world'
```

D.
```
>>> str(divmod(14,9))
'(1, 5)'
```

2.4) Concerning the following program, which statements are incorrect?

```
s1 = '123'
s2 = ' '
s3 = 'abc'

print(s1+s2+s3)
print(s1*2+s2*2+s3)
print(s1*s3)
```

A. The output of the first print() statement is

123 abc √

B. The output of the second print() statement is

123123 abc √

C. The second print() statement will cause an error;

D. The third print() statement will cause an error; √

2.5) Concerning the following program, which of the following statements are incorrect?

```
x=1

if x>2:
    print('Bigger')
    else:
    print('Smaller')

print('Finished')
```

A. This program will output a string ‘Bigger’;

B. This program will output a string ‘Smaller’;

C. This program contains a two-way decision statement;

D. This program will output a string ‘Finished’; √

2.6) Concerning the following two programs:

```
n=5
while n>0:
    print('Lather')
    print('Rinse')
n=n-1
print('Dry off!')
```

```
n=0
while n>0:
    print('Lather')
    print('Rinse')
    n=n-1
print('Dry off!')
```

Which of the following statements are correct?

A. Both programs will NOT cause syntax errors;

B. The while loop in the first program will execute 5 iterations;

C. The while loop in the second program will execute 0 iteration;

D. The output of the second program is:

```
Lather
Rinse
Dry off!
```

2.7) Concerning the following program

```
while True:
    line = input('Input a word:')
    if line[0] == '#': continue
    if line == 'done':
        break
    print(line)
print('Done')
```

Which of the following statements are correct?

A. The while loop in this program is an infinite loop.

B. If the word inputted by the user is 'done', the loop will jump to the next iteration directly.

C. If the first letter of the word inputted by the user is '#', the loop will be terminated directly.

D. The purpose of this program is to print out all the words inputted by the user, except for 'done' and the words starting with '#'

2.8) Concerning for loop and the following program

```
smallest_so_far = None
print('Before',smallest_so_far)

for num in [9,39,21,98,4,5,100,65]:
    if smallest_so_far == None:
        smallest_so_far = num
    elif num < smallest_so_far:
        smallest_so_far = num
    print(smallest_so_far,num)

print('After',smallest_so_far)
```

Which of the following statements are incorrect?

A. for loop is a definite loop in the sense that its iteration number is set beforehand.

B. for loop can only be used to iterate every elements of a list.

C. The purpose of the above program is to find out the smallest number in the list.

D. The value of smallest_so_far is set to None since we are not sure how small the smallest number in that list can be.

2.9) Concerning the following program

```
def greet(lang):
    if lang=='es':
        return 'Hola'
    elif lang=='fr':
        return 'Bonjour'
    else:
        return 'Hello'

>>> print(greet('en'),'Glenn')
Hello Glenn
>>> print(greet('es'),'Sally')
Hola Sally
>>> print(greet('fr'),'Michael')
Bonjour Michael
```

Which of the following statements are correct?

A. lang is an argument of function greet().

B. Function greet() has only one parameter.

C. Function greet() has only one return value.

D. If the input is ‘abc’, function greet() will print out string ‘Hello’ on the screen.

2.10) Concerning the following program

```
numlist = list()
while True:
    inp = input('Enter a number:')
    if inp =='done': break
    value = float(inp)
    numlist.append(value)

average = sum(numlist)/len(numlist)
print('The average is:',average)
```

Which of the following statements are correct?

A. The purpose of this program is to calculate the sum of a list of numbers;

B. This program uses a definite loop to ask the user to input some numbers;

C. The numbers inputted are saved using a list;

D. The elements in numlist have no order;

Question 3. (6% + 14% + 10% = 30%)

Read the following programs and answer the corresponding questions.

3.1) Concerning the following program

0 1 2

```
myText = 'abced*def  abc*sssss!!  123456789'
wordsOne = myText.split()
wordsTwo = wordsOne[1].split('*')
print(wordsOne[2][3])
print(wordsTwo[0])
print(wordsTwo[1][5])
```

('abc', 'sssss!!)

What are the outputs of this program?

4
abc
!

3.2) Concerning the following program

```
wordDict = dict()

myText = "This is a question for CSC1001. And this is a difficult one."
words = myText.split()

print('Counting...')
for word in words:
    wordDict[word] = wordDict.get(word,0)+1

print('The result of word count:')
print(wordDict)
```

Answer the following questions:

Counting... The result of word count:

A. The data type of variable wordDict is? dictionary
B. The data type of variable words is? list
C. What is the functionality of method wordDict.get()? return value from key word
if no key/value provided before → that key has value of 0+1=1
D. What are the outputs of this program?

{'This':1, 'is':2, 'a':2, question:1, for:1, CSC1001:1, And:1, this:1, difficult:1, one:1

3.3) Concerning the following program

```
myDict={'aaa':1,'aab':10,'cde':15,'zoo':200}
mySeq = myDict.items()

newList1 = list()
newList2 = list()

for key,val in mySeq:
    newList1.append((key,val))
    newList2.append((val,key))

print('The list sorted by keys:')
print(sorted(newList1))
print('The list sorted by values:')
print(sorted(newList2))
```

Answer the following questions:

A. How many lists have we defined in this program? 2

B. What are the outputs of this program?

[('aaa', 1), ('aab', 10), ('cde', 15), ('zoo', 200)]

[(1, 'aaa'), (10, 'aab'), (15, 'cde'), (200, 'zoo')]

END OF EXAMINATION